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Physics-Informed Event Attribution From Sparse PMUs Under Model Uncertainty

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ABSTRACT Sparse PMUs can reveal a disturbance without identifying its physical source. We study attribution as a comparison of intervention-conditioned network trajectories, rather than a direct mapping from measurements to bus labels. A frozen differential-algebraic model predicts voltage, terminal current, frequency and ROCOF at eight PMUs. A causal history bank tests load and line hypotheses over a ten-second retrospective window, integrates load severity, retains recovery states after restoration, and combines fixed network-model hypotheses. Its source masses M and increments distinguish conditional evidence from model reweighting. Separate experiments expose the information limit: accurate hidden-voltage recovery does not imply source separation, and weak-pair exact-support Top-1 in the matched batch study is 24.4%, versus 100.0% for finite pairs. In continuous-stream development tests, the physical bank obtains 53/54 timely family/source decisions with 4 unmatched alarms. The nominal reference gains one timely decision but produces more extras; a stricter admission ablation suppresses extras while losing a real intervention. The per-family and temporal analyses explain those outcomes. Confirmatory performance remains pending completion of the frozen campaign. The demonstrated scope is a known load/line catalog, not all event families, unseen-source causal transfer or field operation.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

AN unmeasured load changes. Several PMUs respond, a detector raises an alarm, and the network keeps oscillating after the load is restored. A second intervention then begins during that recovery. The operational question is no longer whether one signal exceeds a threshold: which physical intervention explains the reports observed so far, where did it originate, and when is that explanation distinguishable from the alternatives?

Sparse PMUs make the question difficult because a source need not have a sensor at its own bus. Its voltage, current and frequency effects propagate through the network [1], [2]. Uncertain operating context and network parameters can also change those effects. Moreover, physical OFF is not electrical recovery. A method that resets at restoration can treat the remaining response as another event or lose a real event that begins before the oscillation decays.

Strong existing methods solve parts of this problem. Dynamic state estimators combine power-system equations and PMU histories [3]; supervised classifiers learn sparse-PMU event/source labels [4], [5]; and physical outage detectors compare network hypotheses sequentially [6], [7]. A low state error does not, however, prove source identity. Nor does a high classifier score on represented assets prove source transfer. A useful comparison must distinguish these tasks and keep uncertain explanations visible.

Our approach uses the physical intervention as the object of inference. A load or line candidate changes the network's DAE operator. Starting from an estimated past state, that operator predicts what all eight PMUs should report. Conditional likelihoods compare those responses with the observed history while retaining amplitude, noise and network-model uncertainty. The source/family masses M summarize the competing explanations; their increments separate conditional evidence evolution from reweighting physical models. Retained recovery states compete against new events rather than being discarded at OFF.

The paper makes three scoped contributions:

1) **A physical attribution mechanism:** an executable causal load/line history bank with native terminal-

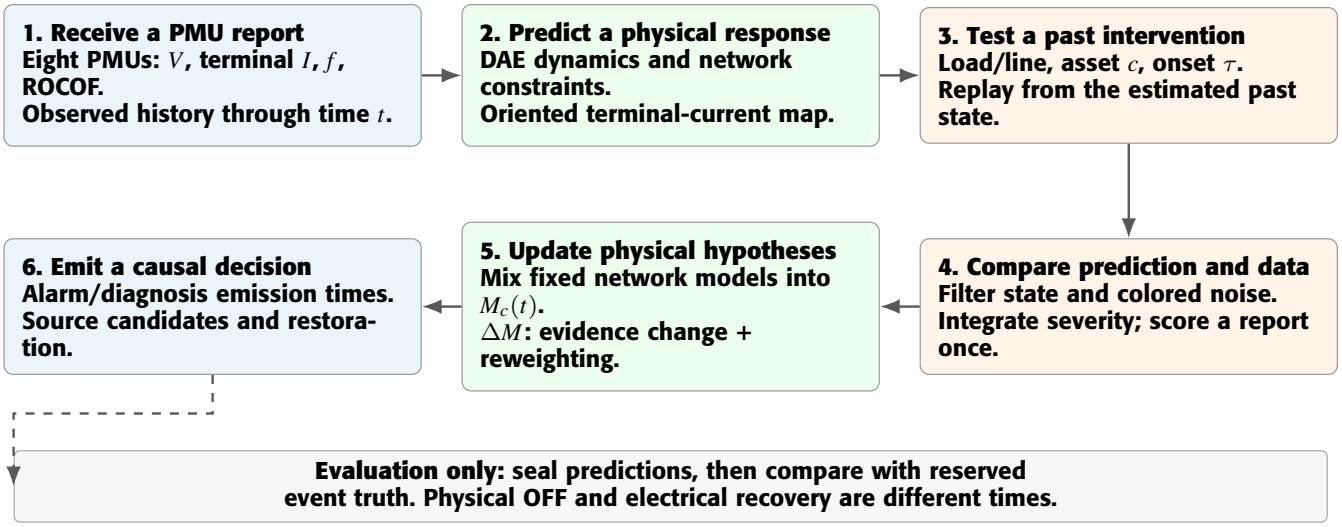


FIGURE 1. **The complete scientific argument. Physics maps an intervention at an unmeasured source into predicted PMU trajectories; causal evidence chooses among those explanations. Event truth enters only evaluation. The figure describes the implemented restricted load/line component, not an already validated all-family joint estimator.**

current predictions, unknown-onset retrospective screening, fixed-model uncertainty and continuous recovery states. We specify the inputs, evidence update and public decisions, not just the architecture.

2) **An information-based interpretation:** functional state recovery and nuisance-profiled source separation answer different questions. The matched load-support study connects this distinction to the observed weak-source resolution limit.
3) **A traceable evaluation argument:** separate state, batch-ambiguity and continuous-stream tests retain their own denominators, negative ablations and comparator eligibility. R11 and nominal are frozen co-references; confirmation is not used to tune another version.

The contribution is not a new Bayes rule, a universally optimal filter, or the name ''tensor'' for an array. It is the physical construction and tested interpretation of competing intervention/recovery explanations under the declared sparse-PMU contract. In completed development evidence, the five-model bank obtains 53/54 timely family/source decisions with 4 extras; nominal retains one more timely event at a higher alarm burden. The per-family breakdown, time-aligned example and failed ablation explain those counts. Final performance claims require the separate frozen confirmation.

II. WHAT EXISTING METHODS EXPLAIN AND WHAT THIS TEST ADDS

Dynamic state estimation supplies the state and uncertainty machinery used to predict a PMU history. Unknown-input and DAE estimators address incomplete models and exogenous inputs [3], [8], [9]. Their task is not automatically an event-family/source decision. A recovered bus-voltage field can remain compatible with competing physical interventions.

Model-based source inference is the closest methodological precedent. Phasor angles support outage hypotheses [10]; uncertain pre-event conditions can be marginalized with outage identity [11]; and sequential likelihoods detect line changes under stochastic operation [6], [7]. We therefore compare with an executable published physical detector rather than presenting physical hypothesis testing itself as new. Its persistent-line assumptions determine where the comparison is eligible.

Learned PMU diagnosis provides a different comparator. Sparse classifiers and knowledge-driven representations identify represented events or assets [4], [5], [12]. The historical SGSMA baseline supplies two frozen ExtraTrees adaptations here. We distinguish interpolation at represented sources, transfer of that frozen adapter, and true held-source generalization. The present causal catalog is known; its results do not establish the last of those claims.

Observability and diagnosability explain why neither a physical nor learned method is guaranteed to localize every visible event. Functional observability concerns the requested state function [13], [14]; descriptor-model diagnosability and nuisance-profiled tests concern competing explanations [15], [16]. Placement-dependent source ambiguity is also established in PMU fault/event work [17]–[19]. These precedents motivate an empirical question, not a claim of first discovery: does preserving the physical event/recovery history remain useful when a fixed sparse sensor set and imperfect network models must support causal decisions?

We use ''physics-informed'' to mean that the physical equations define the candidate trajectories and likelihood inputs. It does not mean that a neural network has been constrained by a physics loss. Learning is present in the supervised references; a learned source-agnostic discrepancy layer and the unrestricted joint state/event/integrity estimator remain separate proposals, not completed contributions.

TABLE 1. The comparison is about inference contracts, not a list of ingredients. The full paper-by-paper literature matrix remains in Supplementary Appendix A.

Route	What drives the decision	Relevant precedent	Role in this paper
State/unknown-input estimation	Dynamic equations and observed history	DAE and recursive estimation	State component; does not alone certify source identity
Learned diagnosis	Features mapped to trained labels	Sparse-PMU classifiers	Frozen causal ExtraTrees references
Physical outage QCD	Likelihood under line hypotheses	Chen *et al.*	Published first-stop comparator on persistent lines
Physical history bank here	Intervention/recovery trajectories with a finite model mixture	Filtering, model selection and diagnosability	Restricted load/line attribution with explicit timing and failure tests

III. THE DECISION REQUIRED FROM EIGHT PMUS

At every report, the task is to decide whether the recent trajectory contains a physical intervention and, if so, identify its family, source and onset. Detection is the existence decision; classification chooses the physical operator; localization chooses a bus or branch. Those outputs need not become reliable at the same time. We keep the estimated physical onset $\widehat{\tau}$ separate from the time at which an alarm or diagnosis is emitted.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$y_k = \mathcal{M}_q(x_k, z_k; \theta_M) + b + \varepsilon_k. \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. The measurement map includes gains, phase references, orientation and the causal frequency/ROCOF estimator. A machine's rotor speed is not substituted for bus frequency.

An intervention hypothesis is $h = (e, c, \tau, a, d)$: family, source, onset, severity and duration. A load changes its local P/Q setpoints. A line opening switches its branch primitive; it is not approximated as a load change. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside C01
Unknown	Unresolved diagnosis	Not a new physical family

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. FROM PHYSICAL PREDICTIONS TO CAUSAL ATTRIBUTION

A. WHERE PHYSICS ENTERS THE ESTIMATOR

The causal bank is the frozen R11 research implementation. Physics determines the candidate response, rather than supplying a penalty to a learned bus-label classifier. Kirchhoff constraints connect each unmeasured source to all available PMUs. With complex bus voltage V, the relevant electrical maps include

$$I_{bus} = Y_{bus} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal-current rows retain transformer ratios, charging and orientation. A terminal current is not the bus injection in the power equation. A line hypothesis changes both the network

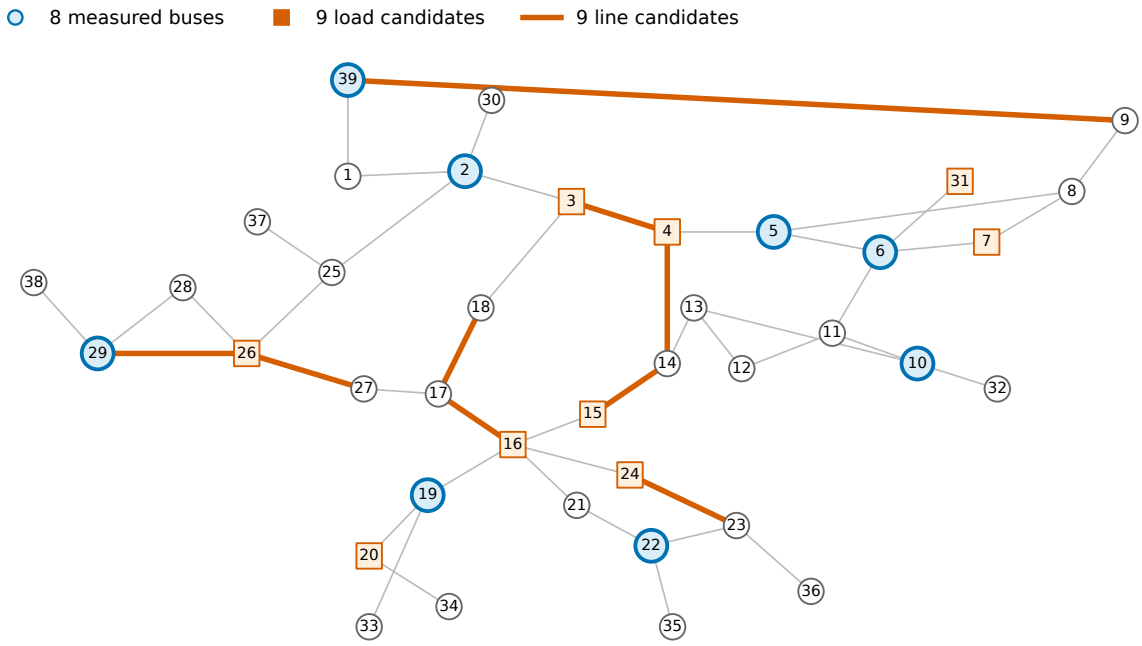


FIGURE 2. **What is measured and what is hypothesized. The frozen causal bank contains nine load buses and nine line assets within the 39-bus network; eight blue buses carry PMUs. An unmeasured candidate is inferred through its network response at those PMUs. Orange branches are candidate interventions, not current-sensor assignments. Coordinates are schematic and do not define electrical distance.**

constraints and the resulting terminal observations; a load hypothesis changes the local input and its dynamic propagation.

On a regular index-one DAE chart, differentiating the algebraic constraint gives the local predictor

$$\begin{aligned}\delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u),\\ \delta\dot{x} &= (f_x - f_z g_z^{-1} g_x)\delta x + (f_u - f_z g_z^{-1} g_u)\delta u.\end{aligned} \tag{6}$$

The frozen implementation builds transition/observation operators from the full DAE derivatives, with native terminal-current dependence for each topology. Equation (6) explains why an input at an unmeasured bus leaves a network-wide signature. It also explains why a local magnitude threshold alone cannot identify the source. Generator inertia, controls and the load law remain those of the frozen plant; no event-specific plant recalibration occurs here.

B. ASK WHAT COULD HAVE HAPPENED IN THE LAST TEN SECONDS

At report time t, screening considers candidate onsets in $[t-10\,\mathrm{s}, t]$. For each source and family it asks: if this intervention began at τ, what reports should have arrived from τ to t? Prediction starts from the stored no-event state estimate at τ, not a reset to nominal equilibrium. This matters when a new intervention arrives during recovery from the preceding one.

For a candidate history, the comparison is between a physical prediction and the same observed window,

$$\begin{aligned} r_{h,W} &= y_W - \widehat{y}_{h,W},\\ L_{h,W} &= \int p(y_W \mid h, a, \mathcal{F}_{\tau^-}) p(a \mid h, \mathcal{F}_{\tau^-})\, da. \end{aligned} \tag{7}$$

Conditional state filtering accounts for the retained physical state and colored instrumental noise. Load amplitude is integrated through the inherited truncated-Gaussian calculation; opening a line uses a switched operator. A finite line-screening response proposes a candidate, and the full switched operator confirms it from the estimated initial state. The rolling screen proposes useful histories; it is not additional independent evidence. A selected suffix is replayed with its original state snapshot, while the sequential likelihood scores each report once.

Supplementary Appendix I gives the scalar precision/mean update and normalizing integral used for a retained load history. It makes the calculation reproducible without presenting standard Gaussian square completion as a new inference theorem.

C. KEEP REFERENCE ERROR SEPARATE FROM EVENT DYNAMICS

A wrong observation center can make a quiet network look disturbed. For each fixed network hypothesis θ_ℓ, startup estimates

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{8}$$

Angular differences use the common reference consistently. Frequency and ROCOF already have empirical startup centers

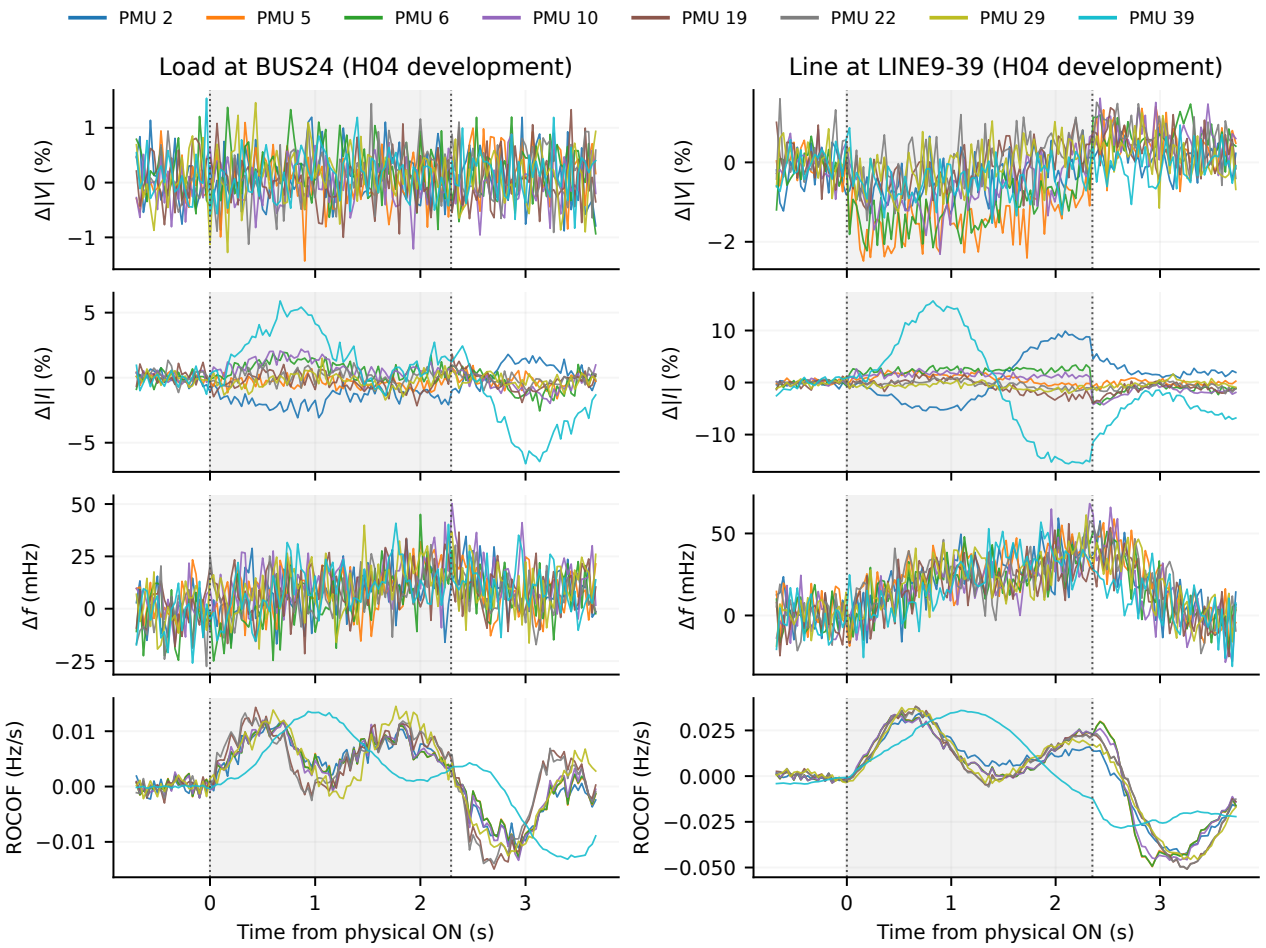


FIGURE 3. **Why the method compares network responses. Two fixed H04 development examples show the first load and first line intervention in parent H002, measured at all eight PMUs. Shading marks the true intervention interval for retrospective explanation only. The different V/I/frequency/ROCOF responses are observations, not universal family templates: source, operating point and severity also change their shape. Frequency and ROCOF include the frozen causal instrumental estimator.**

and receive no extra intercept. The intercept is then frozen: it cannot follow a later load step and erase its signature. This is an observation-discrepancy correction, not recovery of the true impedances or a substitute for an AC-feasible hidden-state estimate. Its uncertainty and the operating-point estimate are treated by substitution.

The preservation mechanism is elementary but useful. For a fixed center, $T_b(y) = y - y^{ref} - b$ satisfies $T_b(y+s) - T_b(y) = s$: a physical response s is not projected away. Updating that center during an intervention would not retain this guarantee. Freezing the center preserves signatures; it does not repair wrong dynamic sensitivities.

Five constant network hypotheses represent nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X changes. Each propagates its own conditional histories. The outer weight uses a predictive density formed before the current-report state update or screening:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{9}$$

Startup reports are not scored again for network-model selection. Pruning, shared covariance approximations and adaptive history proposals make the result an approximate prequential mixture. We do not claim an optimal Kalman filter, a calibrated event posterior or identification of all line parameters.

D. WHAT M AND ´M ACTUALLY DO

Let c index an admissible family/source cell. For a bus load the physical index is diagonal, (i, i, load); for a line it is (i, j, line). The implementation stores only the 18 allowed cells. It does not infer every event at every one of the 39 buses.

For network model ℓ, let $M_{\ell,c,k}$ be the active history mass assigned to cell c. The mixed array and its exact increment

are

$$M_{c,k} = \sum_{\ell} w_{\ell,k} M_{\ell,c,k},$$
$$\Delta M_{c,k} = \underbrace{\sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1})}_{\text{conditional evidence evolution}} + \underbrace{\sum_{\ell} (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}}_{\text{network-model reweighting}}. \quad (10)$$

The array is operational: the mixed masses drive decisions. The decomposition identifies whether an attribution changed within the conditional observers or because network hypotheses were reweighted. It is an accounting identity checked by the code, not a new tensor-factorization theorem. A separate cause mass retains the originating intervention after it switches OFF, so the observer can explain recovery without declaring a new event.

The nominal reference runs the same physical/measurement construction with one network hypothesis and its own publication path. Comparing it with the five-model version tests the value and cost of model uncertainty without changing the plant or its source catalog.

E. EMIT DECISIONS WITHOUT RESETTING THE NETWORK

An alarm, a confirmed family/source diagnosis and an estimated restoration are separate messages. The inherited configuration uses alarm/restoration mass thresholds of 0.99, six confirmation reports for alarm evidence, and 15 reports for restoration confirmation. These numbers are frozen method settings, not calibration guarantees. Additional source/line confirmation and episode-replacement rules remain those of the archived implementation; the configuration is included in the evidence snapshot.

Every emitted message records the latest report used. The observer continues through OFF with its conditional state and covariance; it does not reset to an equilibrium. A later event can begin while the network is still recovering. A source diagnosis emitted after true OFF remains correct eventually but fails the timely-localization metric.

The restricted runtime algorithm is: initialize from the permitted prefix; predict each fixed-model history; score the new report; propose/replay retrospective interventions when supported; mix the history masses; publish a confirmed decision; retain states through restoration. This is model-based Bayesian hypothesis inference. The ExtraTrees comparators are the learned components in this comparison; R11 has no learned bus-label head.

V. WHEN A PHYSICAL EXPLANATION CAN BE DISTINGUISHED

A physical model does not remove an information limit. Two sources can produce nearly the same reports after their amplitudes and initial conditions are allowed to vary. High confidence in the state estimate therefore need not imply high confidence in a source.

The distinction is explicit in a local model. With observation matrix C, transition A and requested hidden-voltage map L, functional reconstruction requires

$$\ker \mathcal{O}_H \subseteq \ker L, \qquad \mathcal{O}_H = \mathrm{col}(C, CA, \ldots, CA^H). \quad (11)$$

Unobservable directions are harmless for this target if they do not change Lx [13], [14]. Source attribution asks a stronger and different question. Let $s_i(\eta_i)$ and $s_j(\eta_j)$ be candidate trajectories after permitting their nuisance coordinates η, including severity and the admitted initial-state variation. Their profiled separation is

$$D_{ij}(H) = \frac{1}{2} \min_{\eta_i, \eta_j} \|s_i(\eta_i) - s_j(\eta_j)\|^2_{R_H^{-1}}. \quad (12)$$

Under a common Gaussian observation law, an intersection gives identical measurement distributions at that intersection. No classifier can then distinguish the candidates uniformly over the allowed nuisance set. A small positive distance can still leave a difficult, noise-sensitive decision [15], [16]. The detailed finite-horizon, signed-support and physical-boundary qualifications remain in Supplementary Appendix D.

These are local conditions, not a structural certificate for the tested IEEE-39 configuration. The archived thresholded Gramian audit has a nonmonotone pseudo-rank and cannot supply that certificate. The nonlinear state-recovery result is evaluated independently; accurate target reconstruction is not used to claim full-state observability.

This framework explains the role of longer history: it may separate response paths that are close in one snapshot, but cannot guarantee an easy decision for every source and severity. It also explains why the estimator must keep competing hypotheses rather than turn every detected change into a confident location.

We test this mechanism with the separate batch reference $\mathcal{E}_{\mathrm{B}}$. It receives 32 rectangular voltage/current coordinates and is given the onset, load family and operating point; it compares 137 zero-, one- and two-source load supports and retains the original GH31 amplitude integration and W2 covariance. It is not R11 with a different solver, and R11 is not silently declared the complete joint estimator $\mathcal{E}_{\mathrm{J}}$. The batch study isolates source ambiguity; the causal study tests sequential decisions under a different contract.

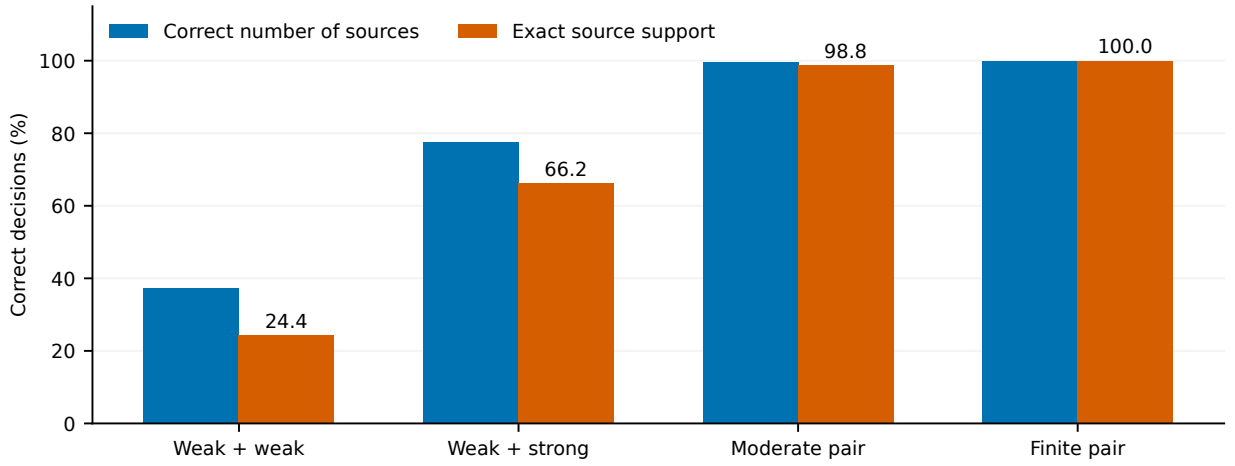


FIGURE 4. **The resolution limit behind the attribution problem. The frozen 137-hypothesis load bank increasingly recovers both cardinality and exact support as event strength increases. Weak-pair exact-support Top-1 is much lower than finite-pair Top-1. Cardinality means choosing the correct number of sources; support means choosing their identities. These known-onset batch results explain ambiguity and do not certify R11's causal performance. Repeated noise decisions are grouped by physical trajectory in the archived protocol.**

VI. EVIDENCE DESIGN AND FAIR COMPARISONS

The experiments answer three successive questions: does the physical model recover useful electrical information; are source responses distinguishable after nuisance fitting; and does a causal decision retain precision when interventions and recovery follow one another? They are separate tests, not parts of one pooled accuracy score.

The causal simulator uses the frozen IEEE-39 Julia/PowerDynamics candidate, with continuous states and a separate PMU reporting grid. A 120-s record contains 3601 reports per PMU; a 600-s normal record contains 18001. Event-aligned reports use the consistent post-switch algebraic value, and solver diagnostics are stored separately. The measurement references, load law and stochastic instrument parameters are not retuned to individual events. RAW0001 is retrospective calibration evidence; RAW0002 remains unopened.

H03 and H04 each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

R11 and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on H04 or C01. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared F01 adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all F01 instrumental effects. We preserve both matched-law implementation checks and F01 mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. It is not assigned a load classifier or a restoration operator. C01 therefore has two evaluation lanes: continuous load/line episodes for R11/nominal/ML, and first-stop persistent-line decisions for all comparators. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

C01 specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three fixed R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories produce 342 intended observed records. All solver and inference failures remain in the ledger. Comparisons use 10,000 paired parent-cluster resamples; conditions and noise realizations do not create extra independent parents. Predictions are sealed before the evaluator opens reserved truth. Until that evaluation exists, the following section reports only completed development evidence.

TABLE 3. One argument, distinct evidence contracts. An event realization repeats a physical parent under a condition or noise draw; it is not another independent physical event.

Question	Executed experiment	Decisive outcome	What it does not establish
Useful state information?	Separate B0–B3 voltage reconstruction and AC recentering	Hidden-voltage error and model-shift sensitivity	Source identity or calibrated joint covariance
Which sources are distinguishable?	Prospective 137-support batch load bank; known onset	Cardinality and exact-support recovery by severity	Unknown-onset multi-family monitoring
Can it monitor successive events?	H03/H04 continuous load/line development	Timely family/source, extras and restoration behavior	Confirmatory superiority or unseen-source transfer
Does the frozen result persist?	C01 configuration sealed before its new predictions	Independent-parent, paired R/X/noise comparison	No result is claimed until execution completes

VII. WHAT WORKS, WHAT FAILS, AND WHY

A. PHYSICAL STATE RECOVERY IS USEFUL BUT DOES NOT LOCATE AN EVENT

The separate matched-model state study reconstructs 31 unmeasured bus voltages with 0.009634% mean relative error. Under operating-point shift, regularized AC-feasible recentering closes 97.4%–98.3% of the tested oracle gap. Yet the nuisance information has rank 25 of 43, and its joint covariance diagnostic remains unreliable. The useful conclusion is target recovery under a stated model, not exact parameter identification or a source-localization guarantee. Full state/mismatch tables and diagnostics remain in the supplement and frozen evidence.

B. PHYSICAL SIGNATURES RESOLVE STRONGER SOURCES, NOT EVERY WEAK PAIR

The batch reference evaluates 137 hypotheses on 3488 fresh nonlinear trajectories. Exact-support Top-1 rises from 24.4% for weak pairs to 100.0% for finite pairs. In the weak regime, even cardinality is imperfect; it nevertheless resolves more often than exact source identity (Fig. 4). This is the distinction in (12): a visible intervention need not have a well-separated location.

The archived retrospective audit associates errors with the closest competing response after amplitude fitting, rather than with a support-only geometric score. The longer-horizon diagnostic increases separation but leaves strongly collinear candidates. Those studies explain a boundary; they do not turn a local numerical condition into a global observability certificate. The original GH31 batch estimator and its denominators remain unchanged.

C. THE CAUSAL GAIN IS IN LOAD RECOGNITION UNDER THIS CONTRACT

The family breakdown in Table 4 is more informative than an aggregate score. In H04, both frozen ExtraTrees adapters and both physical references identify all tested line interventions in time. The physical bank's advantage is concentrated in loads: the five-model version retains 26/27 timely load decisions and its nominal version 27/27, whereas these particular ExtraTrees adapters retain 0/27. This is an observed transfer failure of those frozen adapters on this contract, not proof that supervised learning cannot detect loads. It also cannot be substituted for their historical multi-class hackathon score.

TABLE 4. What the causal component actually resolves in H04 development. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

The five-model physical reference detects and eventually attributes 54/54 realizations, but only 53 diagnoses arrive before physical OFF. Its nominal reference has 54 timely diagnoses. Thus detection, eventual attribution and timely attribution give different answers on the same stream. The nominal reference's additional timely decision comes with more extras: 7 versus 4. Model uncertainty has a measured trade-off, not a uniform benefit.

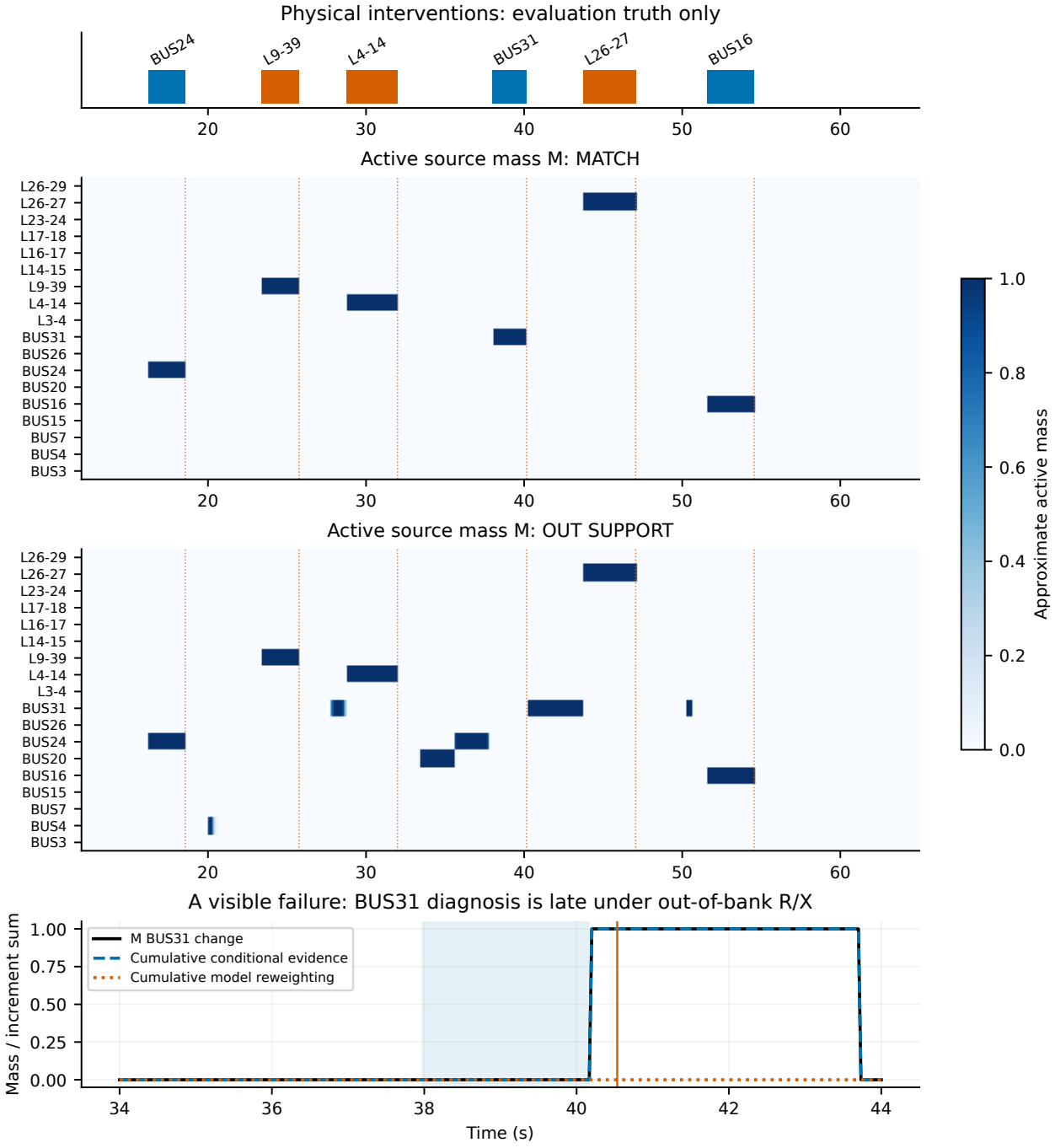


FIGURE 5. **From a sequence to an attribution array. Top: reserved physical intervention intervals in fixed H04 parent H002. Middle: causal active masses across the 18 admissible cells for paired nominal and out-of-bank records. Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 under out-of-bank R/X, with the exact cumulative decomposition of $\acute{M}$ relative to 34 s into conditional evidence and model reweighting. The diagnosis marker is later than OFF: an eventual correct source is not a timely success. This is a development illustration, not C01 confirmation or a calibrated probability display.**

D. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 5 exposes the sequence hidden by a count. A load intervention changes several PMU channels, the history bank accumulates evidence, and the active source mass changes. After OFF, the physical state continues evolving. The retained cause/recovery histories allow the observer to explain that trajectory without resetting the network. A later source must compete against both the no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and (10) separates that revision from within-model evidence. This is what the array adds to the explanation. The figure does not establish that a tensor representation alone improves accuracy.

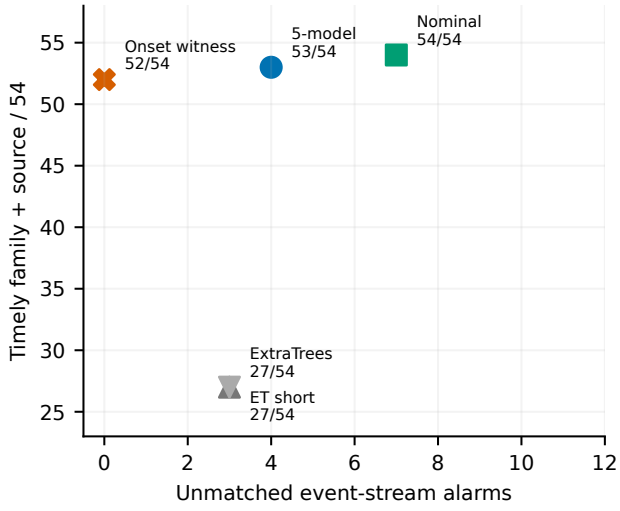


FIGURE 6. The decision that an accuracy-only table hides. H04 development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

An additional onset-witness publication ablation reaches zero extras while reducing timely attribution to 52/54. It rejects a true BUS16 load with weak early onset evidence. The physical states and underlying masses are inherited, so suppressing the public alarm does not repair the estimator's explanation. We retain this as a negative experiment rather than select it for its zero-extra count. R11 and nominal remain frozen for C01.

E. WHAT IS STILL BEING CONFIRMED

The present development evidence supports causal load/line attribution on a known catalog and shows its failure modes. It does not yet support a final performance claim on C01. The confirmatory executor is running with fixed methods, cases, thresholds and metrics; its completed results will enter the dedicated section below through the same hash-to-table/figure chain. No partial C01 result is used here to choose a variant or revise the method.

VIII. INTERPRETATION AND LIMITS

The bank constructs each candidate from the network equations rather than fitting a source-label decision boundary. That is an architectural distinction, not evidence of causal source-excluded generalization. It retains the explanation as states evolve and distinguishes an operator being restored from the grid returning to equilibrium. In this development test the practical gain is timely load attribution; line attribution does not separate the physical and learned references.

That conclusion has three boundaries. First, the causal result uses a known nine-load/nine-line catalog and an intervention-free startup. It is not a test of all 39 buses or arbitrary unknown event families. Second, the instrument and terminal map are frozen research assumptions, including the provisional BUS5 assignment and excluded BUS39 current angle. Agreement within that model does not establish the producer's actual CT placement or exact replication of RAW0001. Third, the network bank is finite and its startup nuisance estimates are plug-in. Its masses are useful attribution weights but are not demonstrated calibrated probabilities.

The nominal reference and failed onset-witness ablation prevent a stronger accuracy claim. More network hypotheses reduce extras in H04 but delay a BUS31 decision; more restrictive admission removes extras but rejects a real BUS16 event. Those are differences in what the decision retains and when it is emitted. Reporting eventual source recovery, or zero extras alone, would select different winners and hide the trade-off.

The mathematical interpretation is equally scoped. Good hidden-voltage reconstruction does not prove event separability. The batch test supports the importance of profiled response overlap, but does not validate every analytic small-amplitude or changing-sensor limit in the supplement. The 137-support batch reference retains its original quadrature; the causal implementation has not been proved to be the unrestricted joint estimator or an exact reduction of it.

The ExtraTrees result tests the particular frozen causal adapters under these physical conditions. Their load failure calls for that restricted interpretation, rather than a general statement that ML cannot recognize loads. Likewise, Chen's colored-noise/DC mismatch is kept visible instead of being used to claim universal superiority over the published algorithm under its own assumptions.

The final decision rests on the prespecified C01 comparison and claim/evidence review, without another bus-specific correction. Faults, generation mechanisms, PMU loss, unseen-source causal transfer and field operation are future scopes, not hidden requirements for reporting this restricted result. The proposed joint estimator remains documented in the supplement so that exploratory work is preserved without being mistaken for the mechanism validated here.

IX. CONCLUSION

Sparse-PMU attribution is a comparison of physical explanations over time. The implemented load/line bank predicts measurements from intervention-conditioned DAE histories, retains recovery states, and mixes fixed network hypotheses. M records the resulting source masses; ΔM separates within-model evidence from changes in model weight. These operations make the decision traceable without turning standard Bayesian or filtering identities into novelty claims.

The completed development evidence shows both utility and a limit. Physical state recovery can be accurate while weak event sources remain ambiguous. In causal H04 development, the five-model reference obtains 53/54 timely joint decisions with 4 extras; nominal improves timeliness at a higher alarm burden. The family and temporal breakdowns explain that result, including the BUS31 delay and the negative BUS16 admission experiment. Final performance claims depend on the frozen confirmatory results. No broader all-family, unseen-source or field claim follows from these tests.

SUPPLEMENTARY MATERIAL

The supplement preserves detailed proofs, electrical and instrumental conventions, the original batch estimator and joint-estimator proposals, expanded component studies, and complete evidence contracts. The pre-revision manuscript is archived in the repository. Empirical data and inference baselines are unchanged.

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